

Light inducible protein degradation in *E. coli* with LOVtag

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Abstract

Molecular tools for optogenetic control allow for spatial and temporal regulation of cell behavior. In particular, light controlled protein degradation is a valuable mechanism of regulation because it can be highly modular, used in tandem with other control mechanisms, and maintain functionality throughout growth phases. Here, we engineered LOVtag, a protein tag that can be appended to a protein of interest for inducible degradation in *Escherichia coli* using blue light. We demonstrate the modularity of LOVtag by using it to tag a range of proteins, including the LacI repressor, CRISPRa activator, and the AcrB efflux pump. Additionally, we demonstrate the utility of pairing the LOVtag with existing optogenetic tools to enhance performance by developing a combined EL222 and LOVtag system. Finally, we use the LOVtag in a metabolic engineering application to demonstrate post-translational control of metabolism. Together, our results highlight the modularity and functionality of the LOVtag system, and introduce a powerful new tool for bacterial optogenetics.

eLife assessment

This paper describes **useful** tools for the light-induced degradation of proteins in bacteria, which will be appreciated by the community. The methods and data analyses are **solid** but the strength of evidence for the tool working as advertised is still **incomplete**. The study will be of interest to colleagues in the fields of microbiology and synthetic biology.

Introduction

Currently, the most widely used optogenetic systems in bacterial synthetic biology rely on transcription to modulate gene expression. Light responsive transcription factors allow genes of interest to be controlled modularly without having to re-engineer promoters. These tools have provided a valuable means to probe regulatory networks (Dessauges et al., 2022; Harrigan et al., 2018; Lugagne and Dunlop, 2019; Olson et al., 2014). However, these networks can be governed by complex control, where post-transcriptional and post-translational

mechanisms work in concert with transcriptional regulation (Chubukov et al., 2014; Link et al., 2013; Mahmoud and Chien, 2018). Light responsive regulators that act beyond transcription are necessary to more fully mimic natural biology and have the potential to improve upon current deficits of transcriptional control. In synthetic contexts, transcriptional regulation suffers from limited response dynamics. In the case of gene activation, the time delay inherent to transcription and translation, which reaches steady-state after >1.5 hours for inducible promoters (Golding et al., 2005; Olson et al., 2014), restricts the speed at which an input signal can be actuated. More problematically, in the case of gene deactivation, decreasing protein abundance is limited by cell division-based dilution or natural protein half-lives, which are in the range of 5-20 hours for the majority of proteins in *E. coli* (Maurizi, 1992). When cells are growing in exponential phase the effective degradation rate of these long-lived proteins is the cell cycle time, which is on the order of 0.5 hours. However, at high cell densities, such as during stationary phase, slow growth rates result in protein half-lives on the scale of 10s of hours (Maier et al., 2011). Subpopulations of cells in stationary phase have been reported to proliferate, however these growing subpopulations are small compared to non-growing cells and add to heterogeneous protein half-lives within the population (Jöers et al., 2020), making it challenging to reliably remove these proteins. It should be noted that a proportion of proteins are actively degraded during slow growth, but these represent the minority in *E. coli* (Gupta et al., 2022). This is a critical issue for metabolic engineering where chemical production is typically carried out at stationary phase in two-stage fermentations (Lalwani et al., 2018).

Post-translational optogenetic control has the potential to address some of the shortcomings of transcriptional regulation because post-translational control mechanisms can function independently of growth-based dilution. Existing approaches to post-translational optogenetic control include the use of light inducible dimers in split protein systems (Baumschlager et al., 2017; Kawano et al., 2015; Nihongaki et al., 2015), domain insertions with light controlled allosteric domains (Dagliyan et al., 2016; Gil et al., 2020), and membrane-confined functional control (Strickland et al., 2012; Wang et al., 2016). These approaches, however, suffer from a lack of modularity, which is a key benefit of the transcriptional control approaches. Split protein systems and domain insertions require extensive engineering for each protein of interest and solutions for a given protein are unlikely to map to proteins with different structures. Likewise, localization-based approaches are only applicable to a small subset of proteins that have location dependent function, such as transcription factors that require nuclear localization (Yumerefendi et al., 2018).

Protein degradation offers potential as a modular post-translational mechanism of control. Endogenous proteolytic machinery is necessary for proteome homeostasis and acts as a global regulatory system (Mahmoud and Chien, 2018). Targeted proteolysis has proven useful in many synthetic biology applications in both eukaryotic and prokaryotic organisms (Andersen et al., 1998; Cameron and Collins, 2014; McGinness et al., 2006; Morreale et al., 2022; Trauth et al., 2019). In eukaryotic cells, light dependent protein degradation has been demonstrated using various mechanisms (Bonger et al., 2014; Deng et al., 2020; Liu et al., 2020; Renicke et al., 2013; Xue et al., 2019). For example, Bonger *et al.* utilized the LOV2 domain of *Avena sativa* phototropin 1 (AsLOV2), to achieve light dependent protein degradation in mammalian cells (Bonger et al., 2014). LOV2 is a blue-light responsive protein that is widely used in optogenetic tools (Pudasaini et al., 2015). LOV2 contains a core Per-Arnt-Sim (PAS) domain surrounded by N- and C-terminal α -helices. Upon blue light illumination, the LOV2 protein undergoes a conformational change where the C-terminal α helix reversibly unfolds and becomes unstructured (Halavaty and Moffat, 2007; Harper et al., 2003; Yamamoto et al., 2009). Incorporation of a degradation-targeted peptide sequence into the C-terminal resulted in light dependent protein degradation in yeast (Renicke et al., 2013).

However, the bacterial proteasome differs from the eukaryotic proteasome in many ways (Mahmoud and Chien, 2018; Schrader et al., 2009). The bacterial proteasome includes several proteases with divergent targeting behaviors and does not utilize ubiquitin as a generalized modification to trigger degradation (Finley, 2009). In bacteria, protein targeting for degradation is predominantly dependent on primary amino acid sequence as opposed to a ubiquitin-like appendage and, apart from certain well-studied cases, the rules governing sequence recognition of bacterial degrons are not fully understood (Baker and Sauer, 2006; Striebel et al., 2009). Because of these complexities, light dependent degradation systems for bacteria have lagged behind their eukaryotic counterparts, where interaction of a ubiquitin ligase with a defined degron can be used as a control mechanism (Bondeson et al., 2022). Nevertheless, optogenetic degradation remains a key target for bacterial synthetic biology applications. In principle, an optogenetic degradation system could be fast-acting, modular, and interface with endogenous proteasome machinery. These features would provide a straightforward way of adding dynamic control to any protein of interest, complementing existing transcriptional tools for optogenetic control of gene expression. One recent system developed by Komera *et al.* utilizes a light responsive split TEV protease to expose or remove constitutively active degradation tags in *E. coli* (Komera et al., 2022). Although this system achieves protein degradation in response to light, it acts indirectly through activation of an exogenous protease. A simpler approach where the degradation tag itself is light responsive would streamline this by eliminating the need for multiple exogenous components.

Here, we develop LOVtag, a modular protein tag based on the AsLOV2 protein that is conditionally degraded in response to blue light in *E. coli*. We demonstrate that attaching this tag to a protein of interest confers light dependent protein instability. We show the modularity of LOVtag by incorporating it into multiple proteins with widely varying function, converting them all into optogenetically controlled systems without the need for extensive protein engineering. In addition, through photocycle stabilizing mutations, we create a version of the LOVtag that responds to infrequent exposure to blue light. We also demonstrate that our degradation tag can be used in concert with other optogenetic systems for multi-layer control by using EL222, a blue light responsive system for transcriptional control, together with the LOVtag. Lastly, we incorporate optogenetic degradation into a metabolically engineered strain to control production of octanoic acid. Overall, this work introduces a new bacterial optogenetic tool that overcomes the drawbacks of transcriptional control by providing post-translational degradation, while avoiding the substantial protein engineering required for alternative post-translational control mechanisms.

Results

Design and characterization of the AsLOV2-based degradation tag

The *E. coli* proteasome consists of five AAA⁺ proteases and is continuously active, either degrading misfolded proteins for quality control or balancing regulatory protein levels (Mahmoud and Chien, 2018). We set out to exploit endogenous proteasome activity in order to design a light responsive protein tag. To do this, we took insight from studies related to native protein quality control. In bacteria, peptides from stalled ribosomes are targeted for degradation through interaction with a tmRNA that appends a short amino acid sequence, known as an SsrA tag, to the C-terminal end of the incomplete protein (Keiler, 2015). The *E. coli* SsrA tag has been studied extensively and is known to interact with the unfoldases ClpX and ClpA (Flynn et al., 2001; Gottesman et al., 1998). Addition of the SsrA peptide sequence to exogenous proteins targets them for degradation by the host proteasome. SsrA mediated

degradation has proven useful in synthetic gene circuit function and biochemical production (Elowitz, 2000; Gurbatri et al., 2020; Stricker et al., 2008; Torella et al., 2013; Ye et al., 2021). A recent structural study of ClpX interacting with the SsrA tag from Fei *et al.* demonstrated that in order for ClpX to unfold a tagged protein, the C-terminal tail needs to be unstructured and sufficiently long to fit into the ClpX pore (Fei et al., 2020). We reasoned that the mechanism of AsLOV2, in which the C-terminal Ja helix becomes unstructured upon blue light absorption, could be utilized to provide light inducible protein degradation.

Biochemical studies have probed the amino acid sequence of the SsrA tag and its role in degradation targeting. Flynn *et al.* demonstrated that ClpA and ClpX unfoldases interact with overlapping residues within the SsrA sequence and that the last three amino acids (L-A-A) are particularly important for successful degradation (Flynn et al., 2001). Importantly, they showed that mutation of the leucine in the C-terminal 'L-A-A' lowers unfoldase affinity but does not hinder degradation completely. We noticed that the C-terminal amino acid sequence of the AsLOV2 domain, comprised of residues 404-546 of *Avena sativa* phototropin 1, are 'E-A-A' at positions 541-543 (Fig. 1a). The dark state structure of AsLOV2 (PDB: 2V1A) shows that these three amino acids and K544 complete the folded Ja helix (Fig. 1b). A truncation of residues 544-546 leaves the Ja helix largely intact and the resulting C-terminal 'E-A-A' remains caged as part of the folded helix. We hypothesized that this truncation would be stable in the dark state as a consequence of 'E-A-A' caging and unstable in the light state due to Ja helix unfolding and exposure of an unstructured degradation tag.

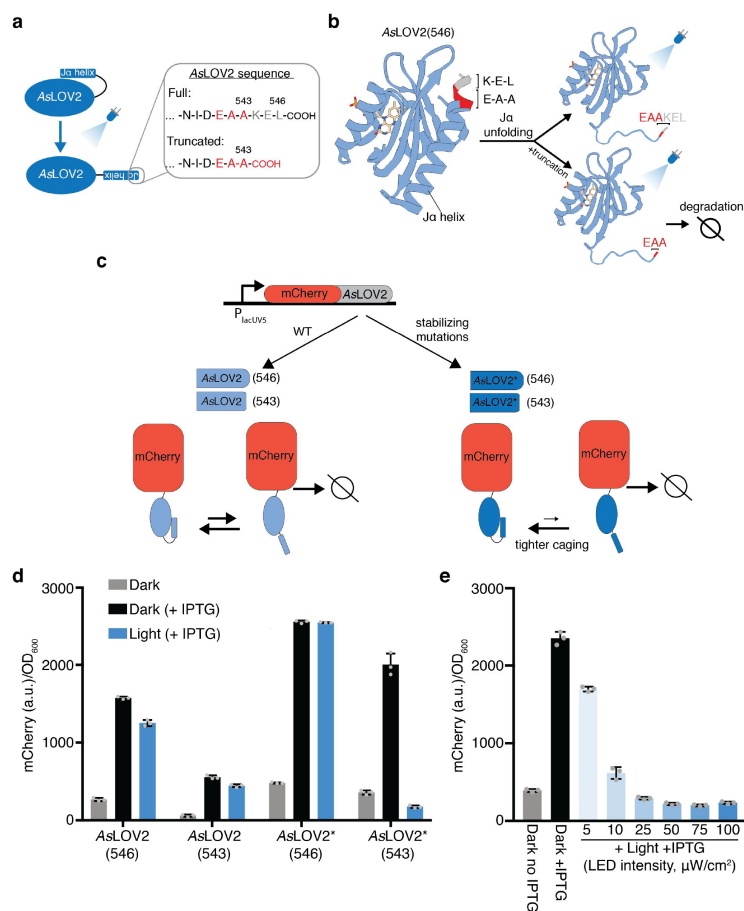


Figure 1.

Design of AsLOV2-based degradation tag. (a) Primary sequence of AsLOV2(546) C-terminal sequence. A three amino acid truncation exposes E-A-A. **(b)** Structure of AsLOV2 (aa404-546, PDB: 2V1A). Amino acids 541-543 (E-A-A) are red and 544-546 (K-E-L) are gray at the C-terminal of the Ja helix. **(c)** Construct used to characterize optogenetic control using AsLOV2 variants. Each variant is translationally fused to mCherry expressed from an IPTG inducible promoter. Variants include wild type AsLOV2 (light blue) and a dark state stabilized version, AsLOV2* (dark blue), with and without the three amino acid truncation. **(d)** mCherry expression levels in response to 465 nm blue light for wild type AsLOV2 and mutated AsLOV2* fusions with and without truncation. **(e)** Expression of mCherry-LOVtag in response to variable light intensities. Error bars show standard deviation around the mean (n = 3 biological replicates).

Optogenetic systems have used the AsLOV2 domain in *E. coli*, however, it is often appended N-terminally or internally to another protein (Li et al., 2022; Strickland et al., 2008). To the best of our knowledge, wild type or truncated AsLOV2, with its C-terminal end exposed, have

not been used for proteolytic degradation in bacterial cells. To test the stability of C-terminal AsLOV2 in *E. coli* in response to light, we constructed a plasmid where we used an IPTG-inducible promoter to control the expression of mCherry translationally fused to AsLOV2 (Fig. 1c). In this construct, the full C-terminal sequence is intact so unfoldases are not expected to have good access for protein degradation. Consistent with this, induction of mCherry-AsLOV2(546) with IPTG increased expression of the fusion construct, and 465 nm blue light induction resulted in only a modest decrease in expression (Fig. 1d). Next, we tested a version of the construct where three amino acids were truncated to expose the C-terminal 'E-A-A', AsLOV2(543). This construct destabilized the protein fusion as predicted, resulting in significantly lower expression compared to the non-truncated version (Fig. 1d). However, counter to our expectations, we observed only a modest decrease in expression upon blue light induction.

While AsLOV2 switches from mostly folded in the dark state to mostly unfolded in the light state, both states are present at an equilibrium with or without light (Yao et al., 2008). Due to the suboptimal equilibrium of native AsLOV2, dark state undocking and unfolding of the Ja helix is a common issue in AsLOV2-based optogenetic tools, and many studies have aimed at improving the dynamic range of AsLOV2 (Guntas et al., 2015; Lungu et al., 2012; Strickland et al., 2012; 2010; Wang et al., 2016). Guntas *et al.*, performed phage display on an AsLOV2-based protein that incorporates a caged peptide sequence in the Ja helix. They identified a variant with 11 amino acid substitutions with much tighter dark state caging, called iLID. Interestingly, several of these mutations do not have a direct interaction with the caged peptide, but instead are located at the hinge loop connecting the PAS domain to the Ja helix or in the PAS domain itself (Fig. S1). Guntas *et al.* further characterized amino acid substitutions and determined several that did not affect caging or dynamic range if reverted (F502Q, H521R, and C530M). We focused on the mutations found to result in tighter dark state caging (L493V, H519R, V520L, D522G, G528A, E537F, N538Q, and D540A), and reasoned that this set should stabilize the Ja helix irrespective of the caged peptide. Thus, we next constructed a variant of AsLOV2 with these eight mutations, which we denote as AsLOV2*.

To test whether these mutations are beneficial in the context of protein stabilization, we fused AsLOV2*(546) or AsLOV2*(543) to mCherry (Fig. 1c). Non-truncated AsLOV2*(546) expressed highly with the addition of IPTG, but blue light did not result in a decrease in mCherry levels, consistent with lack of access by proteases (Fig. 1d). In contrast, the truncated version containing the mutations, AsLOV2*(543), displayed the desired light induced degradation, expressing highly in the dark and exhibiting 11.5x lower expression in response to blue light (Fig. 1d). We also confirmed that AsLOV2*(543) expression was decreased in response to light without any IPTG induction (Fig. S2). Further, AsLOV2*(543) is destabilized in proportion to light intensity (Fig. 1e). We conducted experiments to identify the mechanism underlying AsLOV2*(543) degradation and found evidence that ClpA is involved, and other proteases and unfoldases play complementary roles (Supplementary Text, Fig. S3-6). A potential concern is that light induced disorder of the Ja helix could result in a decrease in solubility and aggregation of mCherry-AsLOV2*(543). To rule this out as the cause of the fluorescence decrease, we captured microscopy images of cells in dark and light conditions. The imaging confirmed a light-dependent decrease in mCherry expression without the formation of visible protein aggregates (Fig. S7). Thus, the AsLOV2*(543) variant provides blue light dependent protein degradation; we refer to this variant as the LOVtag.

Modularity of the LOVtag

Post-translational control of protein function typically requires extensive protein engineering for each use case. Degradation tags, by contrast, offer post-translational control that theoretically requires little to no protein engineering and is protein agnostic. To test the modularity of the LOVtag, we incorporated optogenetic control into three systems with

highly diverse functions and relevance to synthetic biology and biotechnology applications: the LacI repressor, CRISPRa activation, and the AcrB efflux pump.

First, we sought to test whether the LOVtag could be fused to transcription factors to enable light dependent regulation. The LacI repressor is one of the most commonly used chemically inducible systems in synthetic biology. We translationally fused the LOVtag to LacI and paired it with a reporter where the LacUV5 promoter controls expression of mCherry (Fig. 2a). Our results show that light exposure successfully increased mCherry expression (Fig. 2b). Light induced mCherry expression did not achieve the full levels provided with saturating IPTG induction, however we still observed a notable increase. We tested an alternative strategy for further improving de-repression, which suggested that the discrepancy between IPTG versus light dependent induction likely stems from the delay in LacI degradation compared to the rapid allosteric action of IPTG (Supplementary Text, Fig. S8).

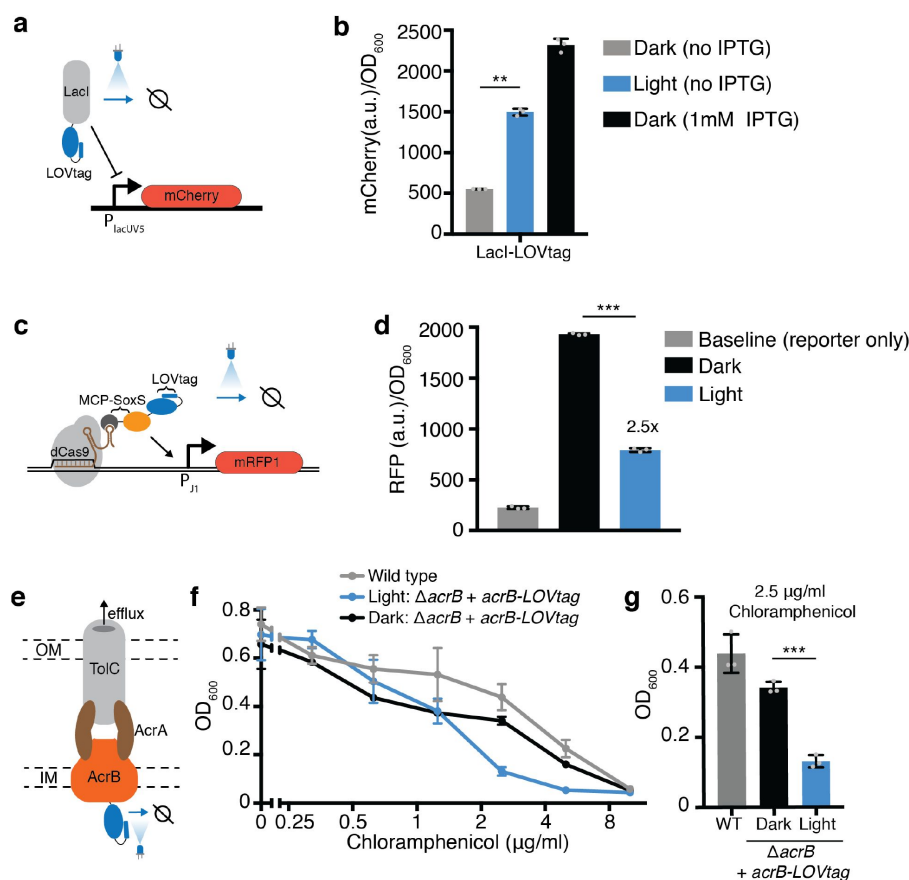


Figure 2.

Incorporating light responsiveness into diverse proteins with the LOVtag. **(a)** Control of mCherry repression using a LacI-LOVtag fusion. **(b)** mCherry expression in response to light exposure for strains with LacI-LOVtag compared to IPTG induction (** $p < 0.001$, two tailed unpaired t-test). **(c)** Schematic of SoxS-based CRISPRa activation with a LOVtag appended to the MCP-SoxS activator. **(d)** CRISPRa control of mRFP1 expression in response to light (*** $p < 0.0001$, two tailed unpaired t-test). **(e)** Schematic of the LOVtag appended to AcrB of the AcrAB-TolC efflux pump. IM, inner membrane; OM, outer membrane. **(f)** Minimum inhibitory concentration (MIC) curves of cells cultured in chloramphenicol. Wild type cells (BW25113) are compared to a Δ acrB strain complemented with AcrB-LOVtag and exposed to light or kept

in the dark. **(g)** OD₆₀₀ of strains shown in (f) at 2.5 µg/ml chloramphenicol (*** $p < 0.0001$, two tailed unpaired t-test). Error bars show standard deviation around the mean ($n = 3$ biological replicates).

Next, we incorporated the LOVtag into the SoxS-based bacterial CRISPRa activation system (Dong et al., 2018). In this system, a scaffold RNA, which is a modified gRNA containing an MS2 stem loop, is used to localize dCas9 and the transcriptional activator SoxS, which is fused to an MS2 coat protein (MCP). We translationally fused the LOVtag to the MCP-SoxS protein, such that in the dark CRISPRa will be active and light exposure relieves activation (Fig. 2c). In the original system, MCP-SoxS expression is anhydrotetracycline (aTc) inducible. This induction system is not amenable to blue light stimulation because aTc is

photosensitive (Baumschlager et al., 2020). Thus, we changed the MCP-SoxS construct to an IPTG inducible promoter prior to blue light experiments (Fig. S9a, Table S1, Table S2). We confirmed that the promoter switch maintained CRISPRa activity (Fig. S9b). Fusing the LOVtag to the activator component indeed relieved CRISPRa activity, resulting in a decrease in expression under blue light stimulation (Fig. 2d).

We also added the LOVtag to the endogenous membrane protein AcrB. This represents a challenging test case for degrading a native protein. The AcrAB-TolC complex is a multidrug efflux pump with clinical relevance due to its role in antibiotic tolerance and resistance acquisition (El Meouche and Dunlop, 2018; Lizarralde-Guerrero and Taraveau, 2021; Okusu et al., 1996). AcrAB-TolC has also been utilized in metabolic engineering as a mechanism to pump out toxic chemical products and boost strain performance (Dunlop et al., 2011; Fisher et al., 2014). However, inducible control of AcrB is challenging because cell viability is sensitive to overexpression (Turner and Dunlop, 2015). Optogenetic transcriptional control systems have high dynamic ranges but do not operate in the low expression ranges relevant to very potent proteins such as AcrB. Light based degradation is well suited for this challenge because it works by decreasing expression, allowing the upper bound of expression to be determined by the promoter.

Previous work from Chai *et al.* demonstrated that AcrB can be targeted for proteolysis by fusing an SsrA tag to the C-terminal (Chai et al., 2016). Degradation is possible because the C-terminal end of AcrB is on the cytoplasmic side of the inner membrane and can interact with cytoplasmic unfoldases (Du et al., 2014). Therefore, we reasoned that fusing the LOVtag to AcrB would result in light inducible degradation that disrupts activity of the AcrAB-TolC complex (Fig. 2e). To determine if activity of the AcrAB-TolC complex was successfully disrupted by light, we performed a minimum inhibitory concentration (MIC) test using the antibiotic chloramphenicol. AcrAB-TolC is known to increase tolerance to chloramphenicol (Okusu et al., 1996). We transformed a construct containing tagged AcrB into cells lacking the endogenous *acrB* gene ($\Delta acrB$). When kept in the dark, $\Delta acrB$ cells with AcrB-LOVtag showed comparable chloramphenicol tolerance to wild type cells (Fig. 2f-g). Blue light stimulation, in contrast, sensitized the $\Delta acrB$ + AcrB-LOVtag strain to chloramphenicol compared to both wild type and $\Delta acrB$ + AcrB-LOVtag kept in the dark (Fig. 2f-g). This suggests that the LOVtag successfully targets AcrB for degradation in a light dependent fashion in a range that is physiologically relevant. When compared to $\Delta acrB$ without any AcrB complementation, the blue light exposed cells do not reach the same level of chloramphenicol sensitivity, likely due to basal levels of efflux pump expression from AcrB-LOVtag relative to the knockout (Fig. S10).

Tuning frequency response of the LOVtag

Light inducible systems have the potential to respond to the frequency of light exposure. Frequency dependent tools can allow optogenetic circuits to be multiplexed beyond limited wavelength options and add a layer of logic to optogenetic circuits (Benzinger et al., 2022). With added logic operations, optogenetic circuits could perform complex signal processing, analogous to those demonstrated with multiplexed chemically inducible circuits (Shin et al., 2020), while allowing dynamic light inputs. Additionally, higher sensitivity LOVtags would also be useful in bioreactor settings, where poor light penetration into dense cultures is a feasibility concern. With these use cases in mind, we sought to investigate and alter the LOVtag frequency response.

LOV domains utilize a flavin cofactor to absorb light. In the case of AsLOV2, the cofactor responsible for light absorption is flavin mononucleotide (FMN). A cysteine residue in AsLOV2 forms a reversible covalent bond with FMN, which initiates broader conformation change. A full photocycle of AsLOV2 consists of absorption of a photon, covalent bond

formation with cysteine 450, J α helix destabilization and unfolding, decay of the cysteine-FMN bond, and J α helix refolding (Swartz et al., 2001). Previous studies have determined the dynamics of bond formation and the time delay of J α helix refolding. Further, mutations have been found that stabilize or destabilize the light state conformation (Christie et al., 2007; Kawano et al., 2013; Zayner et al., 2013). In the context of the LOVtag, the time needed for the J α helix to refold, known as the reversion time, likely determines degradation characteristics (Fig. 3a). In principle, the time spent in the light state dictates the amount of time the protein is susceptible to degradation and, therefore, would impact the frequency response given variable light inputs.

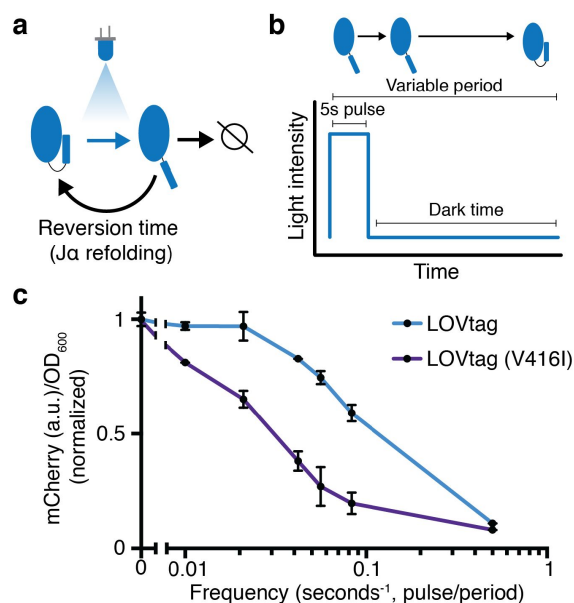


Figure 3.

Modulating LOVtag frequency response with photocycle mutations.

(a) Photocycle of AsLOV2. Upon light absorption, the J α helix unfolds for a period of time dictated by stability of the light state conformation. If not degraded, the J α helix refolds, blocking degradation. **(b)** The light program used to test frequency responses of LOVtag photocycle variants in (c). A constant pulse of 5 sec is followed by a variable dark time that allows for J α helix refolding. **(c)** Expression of mCherry-LOVtag and variant mCherry-LOVtag (V416I) in response to different frequencies of blue light pulses. Fluorescent values are normalized to dark state expression. Error bars show standard deviation around the mean ($n = 3$ biological replicates).

To test the effect of LOVtag refolding dynamics, we illuminated cells with 5 second pulses of blue light followed by variable length dark periods to allow J α helix refolding (Fig. 3b). While holding the blue light duration fixed, we tested dark periods ranging from 475 seconds (5:475 seconds on:off, frequency of 0.01 seconds⁻¹) to 5 seconds (5:5 seconds on:off, frequency of 0.5 seconds⁻¹). We first tested the frequency response of the original LOVtag (Fig. 3c). The response is in line with known refolding dynamics of AsLOV2 (Li et al., 2020), where over 50% degradation is only achieved at high frequencies (0.5 seconds⁻¹, which corresponds to 5:5 seconds on:off). Next, we tested a LOVtag variant that contains a slow-photocycle mutation, V416I (Zoltowski et al., 2009). This amino acid substitution has been shown to increase the dark state recovery from 8 seconds to 84 seconds *in situ* (Li et al., 2020). Indeed, for LOVtag (V416I) over 50% degradation was achieved at medium frequencies (0.04 seconds⁻¹, which corresponds to 5:120 seconds on:off) (Fig. 3c). This variant offers the potential for better performance in settings where increased light sensitivity is preferred, such as bioreactors.

Integrating the LOVtag with EL222 for multi-layer control

Another attractive aspect of post-translational optogenetic control is that it can integrate with existing systems that act at the transcriptional or translational level. Adding control at multiple layers has been shown to enhance the performance and robustness of natural and synthetic systems (Alon, 2007; Hasenjäger et al., 2019; Szydło et al., 2022). One commonly used system for optogenetic transcriptional control is the EL222 protein. EL222 is a blue light responsive LOV protein that dimerizes and binds DNA upon light exposure (Zoltowski et al., 2013). In bacteria, EL222 can be used as a transcriptional repressor or activator depending

on the placement of its binding site in the promoter (Ding et al., 2020; Jayaraman et al., 2016). We chose to combine the transcriptional repression of EL222 with the LOVtag. In this arrangement, the systems work synergistically to decrease gene expression in response to blue light using simultaneous transcriptional repression and protein degradation (Fig. 4a).

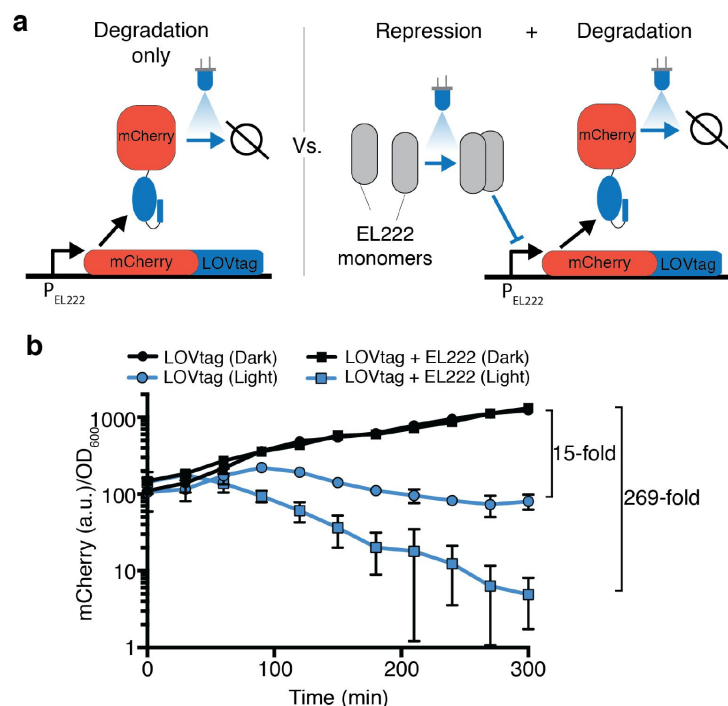


Figure 4.

Enhanced light response using transcriptional control together with the LOVtag. **(a)** The mCherry-LOVtag expressed from an EL222 responsive promoter that is constitutively active in the absence of EL222. Addition of EL222 results in a circuit that both represses and degrades in response to light. **(b)** Light and dark expression of mCherry in the 'degradation only' (circles) or 'repression + degradation' (squares) strains from (a) over time. Error bars show standard deviation around the mean (n = 3 biological replicates).

To test the performance of this combined optogenetic circuit, we created a construct in which the mCherry-LOVtag fusion protein is driven by a constitutively active promoter containing an EL222 binding site (P_{EL222}). We tested the light response of mCherry-LOVtag expression with and without EL222 present (Fig. 4b). As expected, mCherry expression decreased in response to light even without EL222 present, representing the sole action of the degradation tag. However, multi-layer control resulted in a faster decrease in expression in response to light and reached significantly lower levels compared to the degradation tag alone. The fold change decrease in expression was improved from 15x for just the LOVtag to 269x when both systems were combined.

Optogenetic control of octanoic acid production

We next sought to apply the LOVtag to a metabolic engineering task, as we envision that post-translational control will be especially advantageous in these biotechnology applications. Transcriptional control alone is particularly problematic in a metabolic engineering setting because chemical production is typically carried out at stationary phase with slow growth rates, meaning that proteins expressed at basal levels will accumulate in production settings. Therefore, dynamic control using transcriptional optogenetic systems alone will only allow protein levels to increase or plateau. However, a prerequisite for dynamic control of metabolic pathways is that enzyme levels can be modulated to turn off production. Thus, the LOVtag protein degradation system offers excellent potential for control in metabolic engineering applications.

As a proof of concept, we chose to control the enzyme *CpFatB1* with the LOVtag or the EL222-LOVtag circuit. *CpFatB1* is an acyl-ACP thioesterase from *Cuphea palustris* that primarily catalyzes octanoyl-ACP to produce octanoic acid. Octanoic acid is a valuable medium chain

oleochemical with limited natural sources (Sarria et al., 2017). Specifically, we expressed the catalytically enhanced mutant from Hernandez-Lozada, *et al.*, *CpFatB1.2-M4-287*, which we denote here as *CpFatB1** (Hernández Lozada et al., 2018). This enzyme interfaces with the endogenous fatty acid synthesis pathway in *E. coli* to produce free fatty acids (Fig. 5a). In this pathway, the carbon tail of an acyl-ACP moiety is elongated two carbons at a time. *CpFatB1** specifically catalyzes C8-ACPs, which results in production of free octanoic acid. Importantly, since *CpFatB1** is very catalytically active, low levels of expression are optimal for production and strains with high expression exhibit a growth defect (Hernández Lozada et al., 2018). Therefore, to dynamically control *CpFatB1** activity in cells, expression must be controlled in a low range, which is particularly challenging in stationary phase.

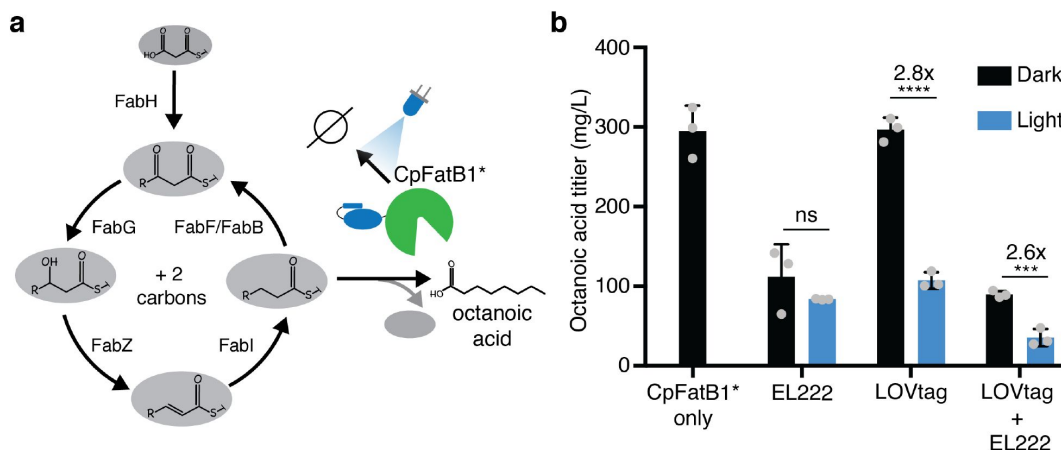


Figure 5.

Optogenetic control of octanoic acid production. **(a)** Schematic of fatty acid synthesis in *E. coli*. *CpFatB1** catalyzes elongating C8-ACP molecules from this pathway to produce free octanoic acid. *CpFatB1** is tagged with a LOVtag to create optogenetic control. **(b)** Octanoic acid titer from strains that express EL222 only, *CpFatB1**-LOVtag only, or *CpFatB1**-LOVtag + EL222 control measured by GC-MS. Strains were kept either in the dark or with continuous blue light exposure for the duration of the production period. Error bars show standard deviation around the mean (*****p* < 0.0001; ****p* < 0.0001; n.s., not significant; two tailed unpaired t-test; *n* = 3 biological replicates).

A first step towards optogenetically controlling this metabolic pathway is to demonstrate that enzyme levels can be modulated between expression levels that are relevant to endpoint titers. If expression can only oscillate between high and medium expression levels, dynamic light control will not be effective. To test whether we could effectively stunt *CpFatB1** activity, we controlled enzyme expression using only EL222 repression, only LOVtag degradation, or EL222 and LOVtag throughout at 24 hour fermentation period (Fig. 5b). The EL222 only control did not significantly decrease octanoic acid production with light, demonstrating the shortcomings of solely transcriptional control at stationary phase. In contrast, both the LOVtag alone and the LOVtag with EL222 resulted in a significant decrease in octanoic acid in the continuous light condition. Thus, protein degradation is needed to effectively shunt the metabolic pathway during stationary phase.

Clearly, decreasing titers is not the overall goal in metabolic engineering. However, the ability to control expression at relevant ranges throughout stationary phase represents an

important stepping stone to investigate dynamic control schemes or feedback control that may lead to enhanced strain performance.

Discussion

Here, we developed and characterized the LOVtag, which provides blue light inducible protein degradation, offering unique advantages as a tool for bacterial synthetic biology. Protein degradation offers a mode of control that is currently limited in the bacterial optogenetic toolkit. The bacterial light responsive degradation system from Komera *et al.* accomplishes protein degradation indirectly through an exogenous split protease (Komera *et al.*, 2022). In comparison, in our approach the degradation tag itself is light responsive and does not require extra components, making it straightforward to incorporate. In this study, we showed that the truncated AsLOV2 protein can be fused to the C-terminal end of various proteins to destabilize them and then target them for degradation. The addition of stabilizing mutations from iLID result in low basal activity in the dark, with strong switching in response to light. Aside from creating a light-responsive degradation tag, the effectiveness of these stabilizing mutations may serve to improve the switch-like behavior in many other AsLOV2-based systems.

We demonstrated the modularity of the LOVtag by incorporating it into three distinct contexts with little to no fine tuning. Each system, the LacI repressor, CRISPRa activation, and the AcrB efflux pump, were converted to optogenetically controlled by simply tagging the protein with the LOVtag. This modularity mimics the ease of use of transcriptional optogenetic systems, which only require a change of the promoter. However, incorporating light control at the protein level simplifies this process because expression levels can remain in their native, or previously fine-tuned, context. For example, the AcrB efflux pump is beneficial to the cell only at low expression levels, as higher expression negatively impacts cell health (Turner and Dunlop, 2015). To maintain low exogenous expression, researchers typically rely on leaky expression from inducible systems or chromosomal integration (El Meouche and Dunlop, 2018; Teelucksingh *et al.*, 2022). Addition of the LOVtag has the benefit that it does not disrupt the system at the transcriptional level and allows protein levels to be decreased further. This configuration is especially beneficial for proteins that require low expression, as transcriptional optogenetic systems often suffer from leaky basal expression. The LOVtag provides a design that can work with endogenous expression levels, circumventing this issue. In this study, we focused on proteins expressed on plasmids from synthetic promoters. However, we anticipate the LOVtag can be useful in systems biology contexts where natural promoters linked to endogenous gene networks are best left untouched. In this case, the LOVtag can add the ability to optogenetically manipulate genes while keeping them in their native gene regulatory context.

Since degradation occurs post-translationally, the LOVtag can easily be integrated with existing optogenetic systems to enhance their function. We demonstrated this by combining the LOVtag with the EL222 repression system, showing that the synergistic action of transcriptional repression and degradation results in a drastic increase in the dynamic range, owing to the much lower off-state under light illumination. In this configuration, EL222 and the LOVtag work coherently to decrease expression. However, the LOVtag can potentially be incorporated into activating systems incoherently to create dynamic functionality, such as pulse generators and inverters (Benzinger *et al.*, 2022).

Throughout this study, we performed experiments using low copy number plasmids that result in moderate expression levels of a given protein of interest. While this is a common use case for synthetic biology, degradation as the sole mode of gene expression control may

be limiting when proteins are at very high expression levels. As an ATP-dependent process that utilizes a finite pool of proteolytic machinery, degradation rates can saturate at sufficiently high expression levels. Furthermore, using protein degradation as the sole mode of gene expression control is wasteful in some contexts. Constitutive transcription and translation of a gene followed by degradation utilizes valuable cellular resources, which is an important consideration in metabolic engineering. To avoid this type of energetic waste, we envision the LOVtag could be used in concert with other modes of regulation, as we demonstrated with EL222. In addition, it is unclear whether the LOVtag is compatible with other prokaryotic (including mitochondrial) proteasomes. Future studies focused on the portability of the LOVtag may address this question and potentially lead to further mechanistic insight.

In summary, the LOVtag offers a straightforward route for introducing optogenetic control of protein degradation in *E. coli*. By lowering the barrier to entry for incorporating light responsiveness into a protein of interest, we envision that systems typically studied with chemical induction or constitutive expression can now be controlled optogenetically without extensive fine-tuning. Furthermore, the LOVtag can act as a circuit enhancer when incorporated into existing optogenetic systems to increase functionality and robustness.

Methods

Strains and plasmids

We used *E. coli* BW25113 as the wild type strain. All knockout strains are from the Keio collection (Baba *et al.*, 2006), which were derived from BW25113. We used golden gate cloning to create all plasmid constructs (Engler *et al.*, 2008) (Table S1). The IPTG inducible constructs were derived from pBbS5c-mRFP1 from the BglBrick plasmid library (Lee *et al.*, 2011). In the constitutive version of the mCherry-AsLOV2 variants, we swapped the IPTG inducible promoter with a constitutive promoter, P_{W7} (5'-ttatcaaaaagagtattgaaataaagtctaacctataggaagattacagccatcgagaggacggcgaa-3'). We used this constitutive version for microscopy and the protease knockout studies; all other experiments used the IPTG inducible promoter.

In all cases of protein fusions to AsLOV2 variants, a five amino acid GS linker of 'S-S-G-S-G' was used between the protein of interest and the LOVtag. In the LacI-LOVtag experiments, pBbS5c-mCherry was also used as the backbone, but with the LOVtag sequence cloned after the *lacI* gene instead of *mCherry*. In the LacI-LOVtag experiment to test the impact of basal LacI, pBbS5c-LacI-LOVtag-mCherry was co-transformed with pBbAa-LacI-Decoy from Wang and Tague *et al.* (Wang *et al.*, 2021). CRISPRa constructs were based on the MS2-SoxS system from Dong *et al.* (Dong *et al.*, 2018). The J106 gRNA sequence from Dong *et al.* was used to target dCas9 upstream mRFP1 under control of the minimal J1 promoter. This gRNA plasmid was constructed using the golden gate assembly method to replace the targeting sequence in the pCD061 backbone (Addgene #113315). The mRFP1 reporter plasmid was derived from pJF076Sa (Addgene #113322) by replacing the ampicillin resistance gene with a kanamycin resistance gene from the BglBrick library. MS2-SoxS-LOVtag was expressed from a variant of pJF093 (Addgene #113323). TetR and its corresponding promoter driving expression of MS2-SoxS were replaced with the LacI-P_{Trc} inducible system from pBbA1c-mRFP1 from the BglBrick library (Lee *et al.*, 2011). The inducer was changed to IPTG because aTc is sensitive to blue light (Baumschlager *et al.*, 2020). The LOVtag was added to this construct with a C-terminal fusion to SoxS. Plasmids containing *acrAB* were built from pBbA5k-*acrAB* from El Meouche *et al.* (El Meouche and Dunlop, 2018). The FLP recombination protocol from

Datsenko and Wanner was used to cure the kanR cassette from the genome of the Δ *acrB* strain (Keio collection, JW0451) (Datsenko and Wanner, 2000).

The slow photocycle variant of the LOVtag, V416I, was constructed using site directed mutagenesis of Valine at amino acid position 416.

EL222 was synthesized by IDT (Coralville, IA) and plasmids were constructed to mimic EL222 repression systems from Jayaraman *et al.* (Table S1) (Jayaraman *et al.*, 2016). A variant of the promoter $P_{BLrep-v1}$ from Ding *et al.*, P_{raB} was used in all EL222 experiments and we refer to it in figures as P_{EL222} (Ding *et al.*, 2020). Plasmid pBbE5k- P_{EL222} -mCherry-LOVtag was co-transformed with pEL222, which constitutively expresses EL222 (Table S2).

For octanoic acid production experiments, the coding sequence of *CpFatB1.2-M4-287* derived from Hernández Lozada *et al.* was synthesized by Twist Biosciences (South San Francisco, CA) and cloned after the P_{EL222} or P_{lacUV5} promoter (Hernández Lozada *et al.*, 2018). Plasmid pBbE5k- P_{EL222} -*CpFatB1**-LOVtag was co-transformed with pEL222 for LOVtag + EL222 production control (Table S2).

Plasmids expressing ClpA, ClpX, and HslU were constructed by amplifying the unfoldase gene from the wild type genome and inserting it into the pBbA8k backbone from the BglBrick library (Lee *et al.*, 2011).

Blue light stimulation

Unless otherwise noted, bacteria were cultured in Luria broth (LB) with appropriate antibiotics for plasmid maintenance at 37 °C with 200 rpm shaking. Antibiotic concentrations used for plasmid maintenance were 30 µg/mL for kanamycin, 100 µg/mL for carbenicillin, and 25 µg/mL for chloramphenicol. All light exposure experiments were carried out with a light plate apparatus (LPA) (Gerhardt *et al.*, 2016) using 465 nm blue light. Overnight cultures of light sensitive strains were diluted 1:50 and precultured in the dark for 2 hours. For IPTG inducible constructs, 1mM IPTG was added when the cells were diluted. After 2 hours in the dark, cells were exposed to blue light in the LPA at a setpoint of 100 µW/cm². Red fluorescence (excitation 560 nm, emission 600 nm) and optical density (OD) readings were taken using a BioTek Synergy H1m plate reader (BioTek, Winooski, VT) after 5 hours of incubation. For CRISPRa experiments, light stimulation was continued for 8 hours prior to RFP and OD readings. For frequency response experiments, the LPA was programmed using its Iris software (<https://taborlab.github.io/Iris/>) to pulse blue light at varying frequencies. A 5 second light pulse was constant for each experiment while the time between pulses was varied (5s, 55s, 85s, 115s, 235s, and 480s).

Minimum inhibitory concentrations

Minimum inhibitory concentration (MIC) experiments for chloramphenicol were performed in M9 minimal media (M9 salts, 2mM MgSO₄, 100 µM CaCl₂) with 1% glucose at 37 °C with 200 rpm shaking. Overnight cultures in LB were initially diluted 1:50 into M9 media for 4 hours. The M9 conditioned cultures were then diluted again 1:20 into 24 well plates containing M9 media with varying levels of chloramphenicol (0, 0.3125, 0.625, 1.25, 2.5, 5, and 10 µg/ml) and grown for 20 hours before measuring optical density (OD) using a BioTek Synergy H1m plate reader (BioTek, Winooski, VT). We also conducted experiments with the Δ *acrB* strain. For these, Δ *acrB* was transformed with pBbA5k-AcrAB-LOVtag (Table S2) and MIC experiments were carried out either in the LPA with constant light illumination or kept in the dark for the duration of growth. No IPTG was added to the Δ *acrB* + AcrAB-LOVtag cultures because basal expression is enough to recover wild type resistance. The same MIC

protocol was performed in the dark using BW25113, $\Delta acrB$ + AcrB, and un-transformed $\Delta acrB$ as controls.

Microscopy

Strains were grown overnight in LB medium. Cultures were refreshed 1:100 in M9 minimal media (M9 salts supplemented with 2 mM MgSO₄, 0.1 mM CaCl₂) with 0.4% glucose for two hours. Samples were then placed on 1.5% low melting agarose pads made with M9 minimal media with 0.4% glucose. Samples were grown at 30°C. Cells were imaged at 100x using a Nikon Ti-E microscope. Blue light exposure was provided by a LED ring (Adafruit NeoPixel 1586), which was fixed above the microscope stage and controlled by an Arduino using a custom Matlab script. Blue light was kept constant except during image acquisitions.

Production experiment

For octanoic acid production experiments, strains expressing *CpFatB1** under various modes of control (Table S2) were cultured in LB overnight with light illumination to maintain low *CpFatB1** expression. Overnight cultures were diluted 1:20 into M9 minimal media with 2% glucose and kept in the light until they reached an OD₆₀₀ of 0.6 unless otherwise noted. The LPA was then programmed to either maintain light for low octanoic acid production or turn off light exposure to induce octanoic acid production for 24 hours prior to fatty extraction and quantification.

Fatty acid quantification

Samples for GC-MS quantification were taken at 24 hours. 400 μ L of vortexed culture was used for fatty acid extraction and derivatization into fatty acid methyl esters as described by Sarria *et al.* (Sarria *et al.*, 2018) with the following minor modifications: An internal standard of nonanoic acid (C9) was added to the 400 μ L sample at a final concentration of 88.8 mg/L and vortexed for 5 sec. The following was then added to the sample for fatty acid extraction and vortexed for 30 sec: 50 μ L 10% NaCl, 50 μ L glacial acetic acid, and 200 μ L ethyl acetate. The sample was then centrifuged at 12,000 g for 10 mins. After centrifugation, 100 μ L of the ethyl acetate layer was mixed with 900 μ L of a 30:1 mixture of methanol:HCl (12N) in a 2 mL microcentrifuge tube. The solution was vortexed for 30 sec followed by an incubation at 50°C for 60 mins for methyl ester derivatization. Once cooled to room temperature, 500 μ L hexanes and 500 μ L water were added to the 2 mL microcentrifuge tube, vortexed for 10 sec, and allowed to settle. 250 μ L of the hexane layer was mixed with 250 μ L ethyl acetate in a GC-MS vial for quantification.

The samples were analyzed with an Agilent 6890N/Agilent 5973 MS detector using a DB-5MS column. The inlet temperature was set to 300°C with flow at 4 mL/min. The oven heating program was initially set to 70°C for 1 min, followed by a ramp to 290°C at 30°C/min, and a final hold at 290°C for 1 min. GLC-20 and GLC-30 FAME standard mixes (Sigma) were tested using this protocol to ensure proper capture of all chain lengths and to gauge retention times. Internal standards were used for quantification, with chain lengths C8-C12 quantified with the nonanoic acid internal standard and C14-C18 quantified with the pentadecanoic internal standard.

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Conflict of Interest

The authors declare no competing interests.

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Reviewer #1 (Public Review):

Specifically controlling the level of proteins in bacteria is an important tool for many aspects of microbiology, from basic research to protein production. While there are several established methods for regulating transcription or translation of proteins with light, optogenetic protein degradation has so far not been established in bacteria. In this paper, the authors present a degradation sequence, which they name "LOVtag", based on iLID, a modified version of the blue-light-responsive LOV2 domain of *Avena sativa* phototropin I (AsLOV2). The authors reasoned that by removing the three C-terminal amino acids of iLID, the modified protein ends in "-E-A-A", similar to the "-L-A-A" C-terminus of the widely used SsrA degradation tag. The authors further speculated that, given the light-induced unfolding of the C-terminal domain of iLID and similar proteins, the "-E-A-A" C-terminus would become more accessible and, in turn, the protein would be more efficiently degraded in blue light than in the dark.

Indeed, several tested proteins tagged with the "LOVtag" show clearly lower cellular levels in blue light than in the dark. While the system works efficiently with mCherry (10-20x lower levels upon illumination), the effect is rather modest (2-3x lower levels) in most other cases. Accordingly, the authors propose to use their system in combination with other light-controlled expression systems and provide data validating this approach. Unfortunately, despite the claim that the "LOVtag" should work faster than optogenetic systems controlling transcription or translation of protein, the degradation kinetics are not consistently shown; in the one case where this is done, the response time and overall efficiency are similar or slightly worse than for EL222, an optogenetic expression system.

The manuscript and the figures are generally very well-composed and follow a clear structure. The schematics nicely explain the underlying principles. However, limitations of the method in its main proposed area of use, protein production, should be highlighted more clearly, e.g., (i) the need to attach a C-terminal tag of considerable size to the protein of interest, (ii) the limited efficiency (slightly less efficient and slower than EL222, a light-dependent transcriptional control mechanism), and (iii) the incompletely understood prerequisites for its application. In addition, several important controls and measurements of the characteristics of the systems, such as the degradation kinetics, would need to be

shown to allow a comparison of the system with established approaches. The current version also contains several minor mistakes in the figures.

Reviewer #2 (Public Review):

In this manuscript the authors present and characterize LOVtag, a modified version of the blue-light sensitive AsLOV2 protein, which functions as a light-inducible degron in *Escherichia coli*. Light has been shown to be a powerful inducer in biological systems as it is often orthogonal and can be controlled in both space and time. Many optogenetic systems target regulation of transcription, however in this manuscript the authors target protein degradation to control protein levels in bacteria. This is an important advance in bacteria, as inducible protein degradation systems in bacteria have lagged behind eukaryotic systems due to protein targeting in bacteria being primarily dependent on primary amino acid sequence and thus more difficult to engineer. In this manuscript, the authors exploit the fact that the J-alpha helix of AsLOV2, which unwinds into a disordered domain in response to blue light, contains an E-A-A amino acid sequence which is very similar to the C-terminal L-A-A sequence in the SsrA tag which is targeted by the unfoldases ClpA and ClpX. They truncate AsLOV2 to create AsLOV2(543) and combine this truncation with a mutation that stabilizes the dark state to generate AsLOV2*(543) which, when fused to the C-terminus of mCherry, confers light-induced degradation. The authors do not verify the mechanism of degradation due to LOVtag, but evidence from deletion mutants contained in the supplemental material hints that there is a ClpA dominated mechanism. They demonstrate modularity of this LOVtag by using it to degrade the LacI repressor, CRISPRa activation through degradation of MCP-SoxS, and the AcrB protein which is part of the AcrAB-TolC multidrug efflux pump. In all cases, measurement of the effect of the LOVtag is indirect as the authors measure reduction in LacI repression, reduction in CRISPRa activation, and drug resistance rather than directly measuring protein levels. Nevertheless the evidence is convincing, although seemingly less effective than in the case of mCherry degradation, although it is hard to compare due to the different endpoints being measured. The authors further modify LOVtag to contain a known photocycle mutation that slows its reversion time in the dark, so that LOVtag is more sensitive to short pulses of light which could be useful in low light conditions or for very light sensitive organisms. They also demonstrate that combining LOVtag with a blue-light transcriptional repression system (EL222) can decrease protein levels an additional 269-fold (relative to 15-fold with LOVtag alone). Finally, the authors apply LOVtag to a metabolic engineering task, namely reducing expression of octanoic acid by regulating the enzyme CpFatB1, an acyl-ACP thioesterase. The authors show that tagging CpFatB1 with LOVtag allows light induced reduction in octanoic acid titer over a 24 hour fermentation. In particular, by comparing control of CpFatB1 with EL222 transcriptional repression alone, LOVtag, or both the authors show that light-induced protein degradation is more effective than light-induced transcriptional repression. The authors suggest that this is because transcriptional repression is not effective when cells are at stationary phase (and thus there is no protein dilution due to cell division), however it is not clear from the available data that the cells were in stationary phase during light exposure. Overall, the authors have generated a modular, light-activated degron tag for use in *Escherichia coli* that is likely to be a useful tool in the synthetic biology and metabolic engineering toolkit.

Reviewer #3 (Public Review):

The authors present the mechanism, validation, and modular application of LOVtag, a light-responsive protein degradation tag that is processed by the native degradosome of

Escherichia coli. Upon exposure to blue light, the c-terminal alpha helix unfolds, essentially marking the protein for degradation. The authors demonstrate the engineered tag is modular across multiple complex regulatory systems, which shows its potential widespread use throughout the synthetic biology field. The step-by-step rational design of identifying the protein that was most dark-stabilized as well as most light-responsive for degradation, was useful in terms of understanding the key components of this system. The most compelling data shows that the engineered LOVTag can be fused to multiple proteins and achieve light-based degradation, without affecting the original function of the fused protein; however, results are not benchmarked against similar degradation tagging and optogenetic control constructs. Creating fusion proteins that do not alter either of the original functions, is often difficult to achieve, and the novelty of this should be expanded upon to drive further impact.